



SIMATS
ENGINEERING



SIMATS
Savitribi Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Computer Vision in Autonomous Vehicles

Enhancing Road Safety through Artificial Intelligence

Presented by

Name : M.JEEVITHA

Register Number : 192424257

Department : Artificial Intelligence & Data Science

Course : DSA0204- Computer Vision With
Open CV



Engineer to Excel



1. CCD vs CMOS Sensors

- A Comparative Study of Image Sensor Technologies



Engineer to Excel

INTRODUCTION & PROBLEM STATEMENT



- **What is Computer Vision?**
- Computer Vision is a branch of Artificial Intelligence (AI) that enables computers to capture, process, and interpret images and videos.
- It helps autonomous vehicles understand their surroundings by recognizing roads, vehicles, pedestrians, and traffic signs.
- The system continuously analyzes visual data to make real-time driving decisions.
- It mimics human vision using cameras, sensors, and deep learning algorithms.
- Computer Vision is one of the core technologies behind self-driving vehicles.
- **Problem Statement**
- Human driving errors cause a majority of road accidents.
- Drivers may become distracted, tired, or fail to react quickly.
- Poor weather and low-light conditions reduce visibility.
- Traditional driving cannot always ensure maximum road safety.



Engineer to Excel



WORKING OF COMPUTER VISION

- **Computer Vision Pipeline**

Camera → Image Acquisition → Image Processing → Object Detection → Decision Making → Vehicle Control

- **Key Steps**

- Cameras continuously capture real-time images of the road.
- Image processing enhances image quality by reducing noise and improving clarity.
- AI detects vehicles, pedestrians, traffic lights, road signs, and lane markings.
- Deep learning algorithms classify detected objects accurately.
- The vehicle decides whether to brake, accelerate, turn, or change lanes.
- Control signals are sent to the steering, braking, and acceleration systems.



Engineer to Excel

APPLICATIONS & BENEFITS



Applications

- **Lane Detection:** Keeps the vehicle centered and prevents lane departure.
- **Traffic Sign Recognition:** Detects stop signs, speed limits, and warning signs.
- **Pedestrian Detection:** Identifies people crossing the road and applies emergency braking.
- **Collision Avoidance:** Detects nearby vehicles and obstacles to reduce accidents.
- **Automatic Parking:** Identifies parking spaces and assists in parking.
- **Driver Monitoring:** Detects driver fatigue and distraction.

Benefits

- Reduces road accidents caused by human error.
- Improves driving accuracy and safety.
- Enables real-time decision making.



Engineer to Excel

CHALLENGES & FUTURE SCOPE



- **Challenges**
- Rain, fog, snow, and poor lighting reduce image quality.
- Complex traffic environments make object detection difficult.
- High computational power is required for real-time image processing.
- AI systems must make safe decisions in unexpected situations.
- **Future Scope**
- Fully autonomous Level-5 vehicles.
- AI-powered smart cities and smart traffic management.
- Better cameras, LiDAR, radar, and sensors.
- Faster processing using Edge AI and 5G networks.

CONCLUSION

- Computer Vision enables autonomous vehicles to perceive and understand their surroundings.
- It improves road safety through accurate object detection and real-time decision making.
- AI-powered vision systems reduce accidents and enhance driving efficiency.
- Despite challenges like weather conditions and computational complexity, the technology continues to improve rapidly.
- Future advancements in AI, sensors, and communication technologies will make fully autonomous transportation a reality.
- Computer Vision is transforming the future of intelligent mobility.



Engineer to Excel



2.Challenges in Image Sensing Systems

- Limitations and Issues in Digital Image Capture



Engineer to Excel

INTRODUCTION & PROBLEM STATEMENT



- **What is Computer Vision in Healthcare?**
- Computer Vision enables computers to analyze and interpret medical images such as X-rays, CT scans, MRI scans, and ultrasound images.
- It helps doctors detect diseases more quickly and accurately.
- AI algorithms identify patterns that may be difficult for the human eye to notice.
- It supports early diagnosis and improves treatment planning.
- It reduces the workload of healthcare professionals and enhances patient care.
- **Problem Statement**
- Manual diagnosis is time-consuming and prone to human error.
- Increasing patient numbers put pressure on healthcare systems.
- Some diseases are difficult to detect at an early stage.
- Delayed diagnosis can reduce treatment success.



Engineer to Excel

WORKING OF COMPUTER VISION IN HEALTHCARE



Computer Vision Workflow

Medical Image → Image Preprocessing → Feature Extraction → Disease Detection → Diagnosis → Treatment Support

Key Steps

Medical images are captured using X-ray, MRI, CT, or ultrasound.

Image preprocessing improves quality by reducing noise and enhancing contrast.

AI extracts important features such as shapes, edges, and abnormal tissues.

Deep learning models classify healthy and diseased regions.

The results assist doctors in making faster and more accurate diagnoses.

Doctors verify AI predictions before planning treatment.



Engineer to Excel



APPLICATIONS & BENEFITS

Applications

- **Cancer Detection:** Detects tumors in mammograms and CT scans.
- **Brain Disorder Detection:** Identifies strokes and brain tumors from MRI images.
- **Eye Disease Detection:** Diagnoses diabetic retinopathy and glaucoma.
- **Bone Fracture Detection:** Detects fractures in X-rays.
- **Skin Disease Analysis:** Identifies skin cancer and other conditions.

Benefits

- Improves diagnostic accuracy.
- Enables early disease detection.
- Supports doctors in decision-making.
- Enhances patient safety and treatment outcomes.



Engineer to Excel



CHALLENGES & FUTURE SCOPE

Challenges

- Large amounts of medical data are required for training AI models.
- Protecting patient privacy and medical records is essential.
- AI predictions depend on image quality.
- AI should support doctors, not replace them.

Future Scope

- AI-assisted robotic surgery.
- Personalized treatment using medical imaging.
- Wearable health monitoring devices.
- Faster disease prediction using cloud computing.
- Smart hospitals with automated diagnosis systems.

CONCLUSION

Computer Vision is transforming healthcare by improving disease detection and medical image analysis.

It supports doctors with faster and more accurate diagnoses.

AI helps reduce medical errors and improves patient care.

Despite challenges such as data privacy and image quality, advancements in AI continue to improve healthcare.

Computer Vision is expected to play a major role in the future of digital medicine



Engineer to Excel



3.Impact of Environmental Conditions on Image Acquisition

- Effects of Lighting, Weather, and Environment on Image Quality



Engineer to Excel



INTRODUCTION & PROBLEM STATEMENT

- **Introduction**
 - Human Vision is the natural ability of the eyes and brain to perceive and understand the surrounding environment.
 - Computer Vision enables machines to capture, analyze, and interpret images using cameras and AI algorithms.
 - Both systems aim to recognize objects and understand visual information.
 - Computer Vision attempts to replicate the capabilities of human vision.
 - It is widely used in automation, healthcare, robotics, and security.
- **Problem Statement**
 - Human vision is limited by fatigue, distractions, and environmental conditions.
 - Machines require intelligent vision systems to perform tasks without human intervention.
 - The challenge is to make Computer Vision as accurate and adaptable as human vision



Engineer to Excel

HOW BOTH SYSTEMS WORK



Human Vision Process

- Eyes capture light from objects.
- The retina converts light into electrical signals.
- Signals are transmitted to the brain through the optic nerve.
- The brain interprets shapes, colors, depth, and motion.

Computer Vision Process

- Cameras capture digital images.
- Images are preprocessed to improve quality.
- AI algorithms extract important features.
- Objects are detected and classified.

COMPARISON & APPLICATIONS

Human Vision	Computer Vision
Uses eyes	Uses cameras
Brain processes images	AI processes images
Learns from experience	Learns from datasets
Can become tired	Operates continuously
Limited memory	Large digital storage

Applications of Computer Vision

- ✓ Facial Recognition
- ✓ Self-Driving Cars
- ✓ Medical Imaging
- ✓ Industrial Automation
- ✓ Security Surveillance
- ✓ Robotics



Engineer to Excel



ADVANTAGES & LIMITATIONS

Advantages of Computer Vision

- Faster image analysis.
- High accuracy with AI models.
- Continuous operation without fatigue.
- Processes large volumes of visual data.
- Supports automation in various industries.

Limitations

- Performance depends on image quality.
- Requires large datasets for training.
- Sensitive to lighting and weather conditions.
- High computational requirements.
- Ethical concerns related to privacy.

FUTURE SCOPE

- Smarter AI-powered vision systems
- Advanced robotics
- Smart cities
- Healthcare automation
- Improved human-computer interaction



Engineer to Excel

CONCLUSION



- Human Vision and Computer Vision both interpret visual information but use different mechanisms.
- Human Vision relies on biological processes, while Computer Vision uses cameras and AI algorithms.
- Computer Vision provides speed, automation, and scalability for real-world applications.
- Although it cannot completely replicate human intelligence, continuous advancements in AI are making vision systems more reliable and efficient.
- The future of Computer Vision lies in creating intelligent systems that work alongside humans in healthcare, transportation, security, and industry



Engineer to Excel



THANK YOU